- *AI model*—We'll use OpenAI's REST APIs to access pretrained models, so we can integrate natural language processing capabilities into Astra AI without building our own model from scratch.
- *Ephemeral storage*—For this first iteration, we'll avoid storing conversation history.
- *Automated testing*—We will provide unit tests using Vitest.
- *Error handling*—The app will gracefully handle unexpected situations and provide informative messages to the user.
- *Security*—While the initial focus is on functionality, we will utilize basic security measures like input validation and sanitization. No user authentication will be implemented in this iteration.

A preconfigured starter project

To make your journey smoother and let you focus on the core concepts, I've created a starter project that you'll use throughout the book. This project, located in the starter-project folder of the GitHub repository, is a preconfigured, clean, Next.js application.

Think of it as the foundation of your house. I've already handled all the initial setup, like configuring Tailwind CSS, setting up basic routing, and including essential files, so you don't have to. You'll simply copy this project at the beginning of each chapter and build the new features on top of it. You can use the starter project as a base for all the examples in this book, ensuring you have a consistent and ready-to-run environment from the very beginning.

For this first project, we'll start with a clean slate. We won't be using a preconfigured starter project or the Vercel AI SDK just yet, so you can head to the next section.

2.2.3 Setting up

Now let's look at the code that implements the application. To run this example, you'll need Node.js, which you can install by following the instructions on the official Node.js page at https://nodejs.org/en/download/package-manager.

> **NOTE** All examples and projects were developed and tested using Node.js v18 (long-term support [LTS]). I recommend using an LTS version of Node.js for stability, such as v18.x.x or the current LTS. While our code is designed to be compatible with newer versions of Node.js (including v22.x.x or v24.x.x "current" releases), sticking to an LTS version helps ensure a consistent development environment.

While you won't need to write any code in this chapter, feel free to download the complete project and run it yourself to gain practical experience.

> **NOTE** The source code for the examples in this chapter is available at https://github.com/Generative-AI-Web-Apps/Code.